



សាកលវិទ្យាល័យភូមិន្ទភ្នំពេញ
ROYAL UNIVERSITY OF PHNOM PENH

Library Management System (Phase 3)

Software Engineering
Professor: Lim Lyheng
Class: M1
Group: 8

Group Member

2



Name	Name in khmer	Role
Sou Chanrojame	ស៊ី ចាន់រ៉ូជ័ម	Leader
Orn Pheakdey	អ័ន ភីក្លី	Follower
Long Neron	ឡុង នីរុន	Follower
Sokkong Somnang	សុខគង់ សំណាង	Follower

Table of Contents

1 Introduction

2 Demonstration Goal

3 Feature

4 Real-world Use Case

5 Demo

Introduction

We are showcasing our **Library Management System**, a digital solution designed to replace manual paperwork with seamless automation. This platform streamlines book tracking, member management, and resource allocation to enhance the overall experience for both librarians and patrons.



Demonstration Goal

The primary objective is to demonstrate how this system **optimizes library operations** through:

- **Efficiency:** Automating check-outs, returns, and cataloging.
- **Accuracy:** Maintaining real-time data on book availability and overdue fines.
- **Accessibility:** Providing a user-friendly interface for easy resource discovery.



Feature

- User Login and Authentication
- Roles and permissions (admin, librarian, borrower)
- Manage librarians, staff, students, and members.
- Add, update, delete, and categorize books.
- Track available, borrowed, reserved, and lost books.
- Search books by title, author, subject, or category.
- Online e-book reading.
- E-book reader can like and comment online book
- Generate reports
- Multi-language support. (Khmer, English, Chinese, Japanese, Korea,etc.)
- AI Assistant (summary book, ask AI)



Real-world Use Case

Academic Institution (University Library)

- **The Problem:** Thousands of students trying to locate specific research journals across multiple floors.
- **The Solution:** The LMS provides a **Digital Catalog (OPAC)** with floor-mapping.
- **Outcome:** Students find resources in minutes via their smartphones, and librarians spend less time directing traffic and more time assisting with complex research.



Public Library System (Multi-Branch)

- **The Problem:** A patron wants a book that is currently only available at a branch 10 miles away.
- **The Solution:** Inter-Library Loan (ILL) tracking.
- **Outcome:** The system automates the transfer request, tracks the book's transit between branches, and notifies the patron when it arrives at their local branch for pickup.



Digital Asset Management (E-Library)

- **The Problem:** Students need access to textbooks or research papers outside of library hours or during remote learning.
- **The Solution:** An integrated **E-Book & PDF viewer** with DRM (Digital Rights Management) controls.
- **Outcome:** 24/7 access to resources without physical wear and tear on books, allowing multiple users to "borrow" digital copies simultaneously.



Demo

